

Pressure Measurement



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- Pressure: Pressure is defined as force per unit area of surface.
- Classification of Pressure:
 - **Absolute pressure** is zero-referenced against a perfect vacuum.
 - **Gauge pressure** is zero-referenced against ambient air pressure.
 - **Differential pressure** is the difference in pressure between two points.

Units of Pressure

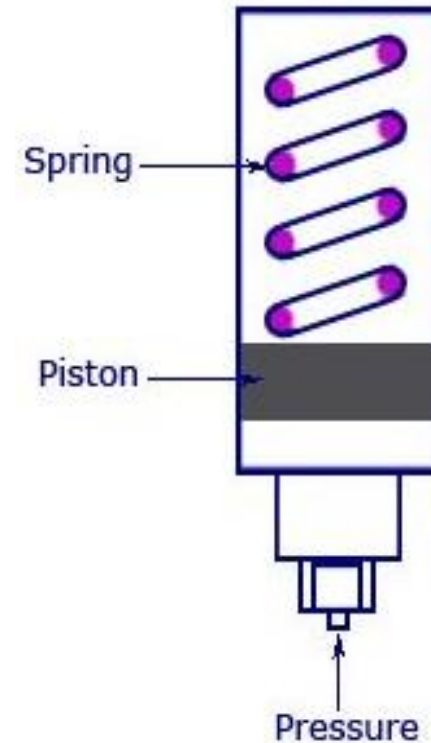
- Pascal-1 N/m² SI unit of Pressure.
- Bar- 100 kN/m²
- atm- 101.33 kN/m²
- Torr- 1 mm of Hg or 0.1333 kN/m²
- Pounds per square inch (psi)- 6.895 kN/m²
 - It is used in USA and Canada.
- psia- Pounds per square inch for absolute pressure.
- psig- Pounds per square inch for gauge pressure.
- psid- Pounds per square inch for differential pressure.

Classification of Pressure measuring Devices

- Hydrostatic Pressure Instrument
 - Piston
 - Liquid Column or Manometer
- Mechanical Pressure Instrument
 - Bourdon Tube
 - Diaphragm
 - Bellows
- Vacuum Pressure Instrument
 - Thermocouple Gauge
 - Pirani Gauge
 - Ionization Gauge

Piston Type pressure measuring device

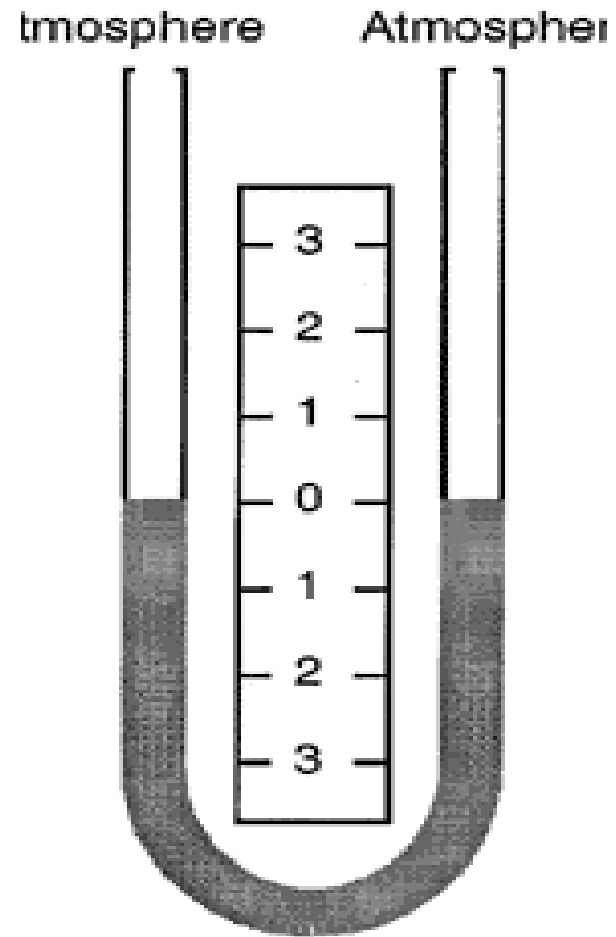
- Piston-type gauges counterbalance the pressure of a fluid with a spring or a solid weight.
- Pressure directly acts on the piston and compress the spring.
- Position of piston is directly related to the pressure.
- A window is made at the outer case to measure the pressure.
- Piston movement can be directly connected to the secondary device to get electrical output.



Piston Type Pressure Transducer

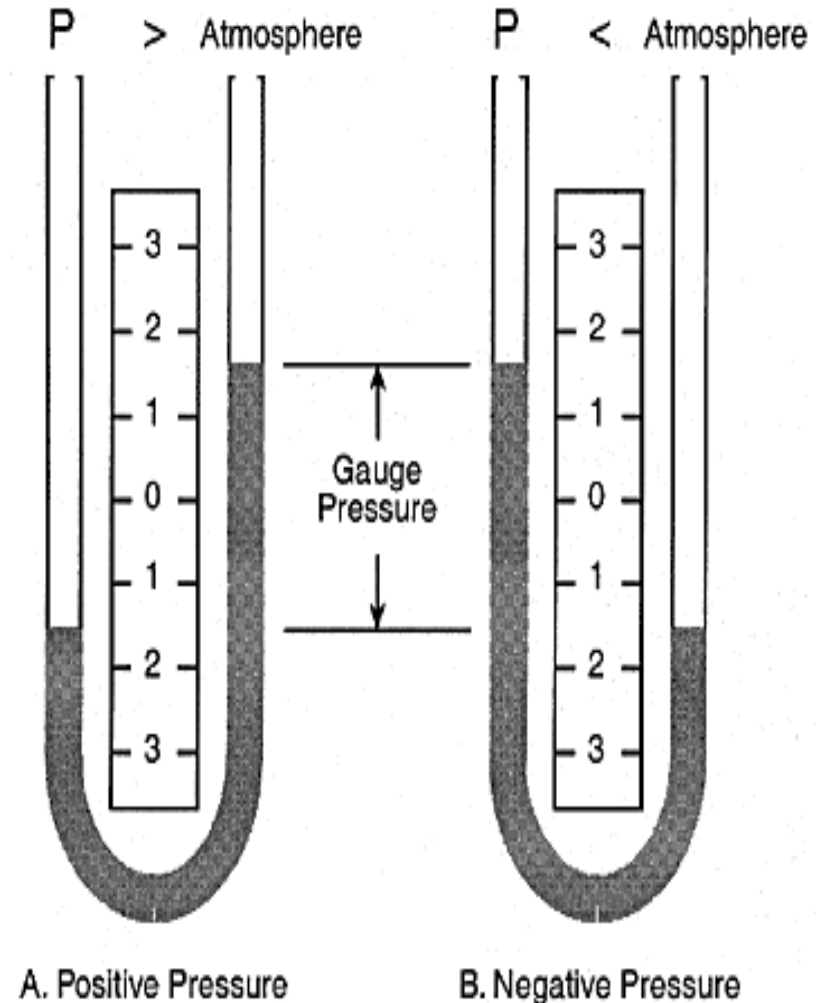
U-Tube Manometer

- U-shaped glass tube partially filled with liquid.
- Water ($\rho = 1000 \text{ kg m}^{-3}$) or Mercury ($\rho = 13560 \text{ kg m}^{-3}$) may be used as liquid in Manometer.
- This manometer has no moving parts and requires no calibration.
- Manometry measurements are affected by gravity and the liquid's density.



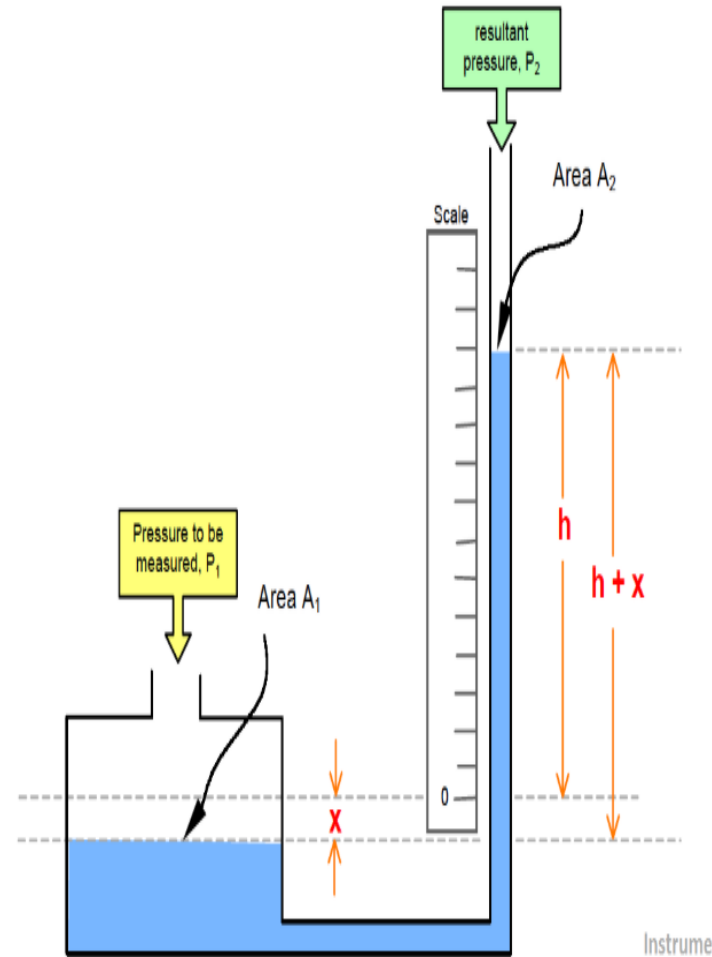
- Each leg of a U-tube manometer exposed to the atmosphere, the height of liquid in the columns is equal.
- Using this point as a reference and connecting each leg to an unknown pressure, the difference in column heights indicates the difference in pressures.
- This Manometer can be used to measure pressure above and below the atmospheric.

$$\Delta p = P_2 - P_1 = \rho gh$$



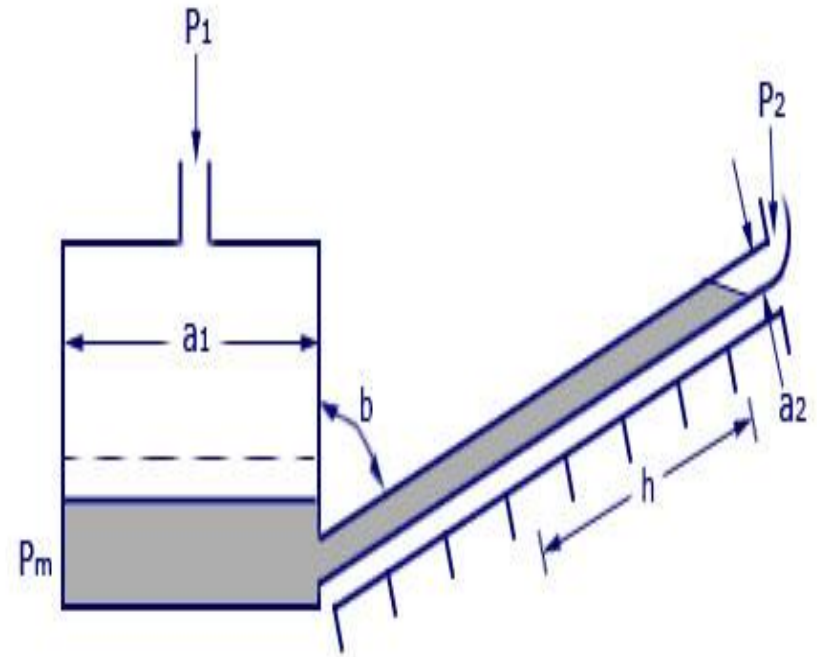
Well-Type Manometer

- It is sometimes called a single column gauge.
- The manometer consists of a metal well of large cross sectional area connected to a glass tube, or limb, filled with mercury .
- In the well type manometer, the pressure to be measured is normally applied to the well.
- The value of (h) will increase as the pressure to be measured increases and will decrease as the pressure to be measured decreases.
- The scale is normally calibrated in units of pressure, e.g. mm of mercury gauge or Pascal (Pa), so that the pressure can be read directly from the device.



Inclined Tube Manometer

- The inclined tube manometer is an enlarged leg manometer with its measuring leg inclined to the vertical axis by an angle b .
- This is done to expand the scale and to increase the sensitivity.
- The factor $\cos b$ expands the scale of the instrument. When b is quite large, h can be increased such that $(h \cdot \cos b)$ remains constant.



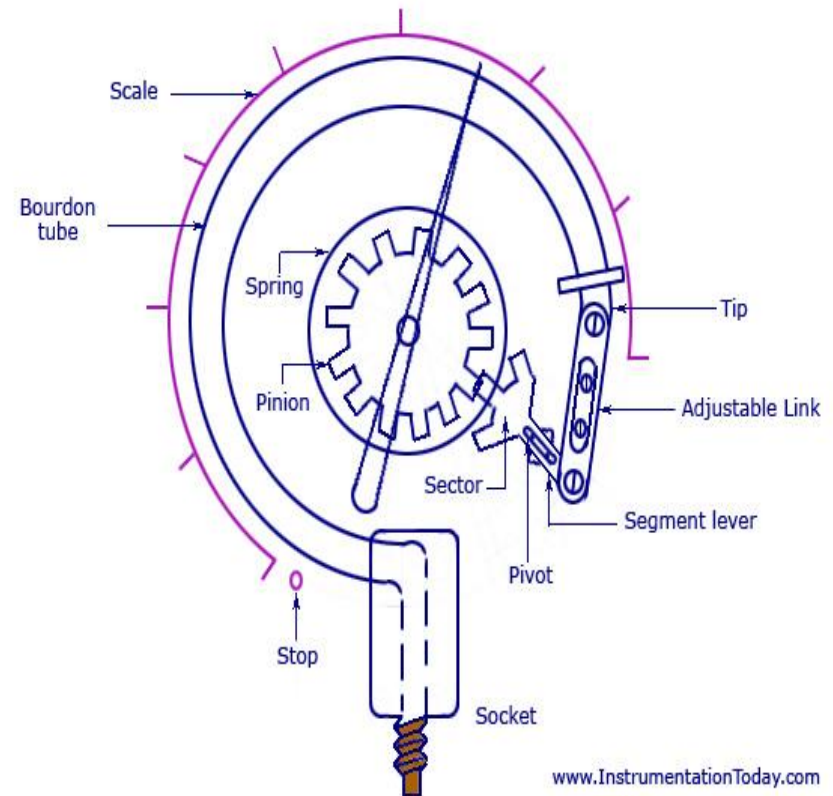
Inclined Tube manometer

Mechanical Pressure Devices

- Bourden Tube
 - C-Type
 - Spiral Type
 - Helix Type
- Diaphragm
 - Flat Type
 - Corrugated Type
 - Capsule Type
- Bellows

C-Type Bourdon Tube

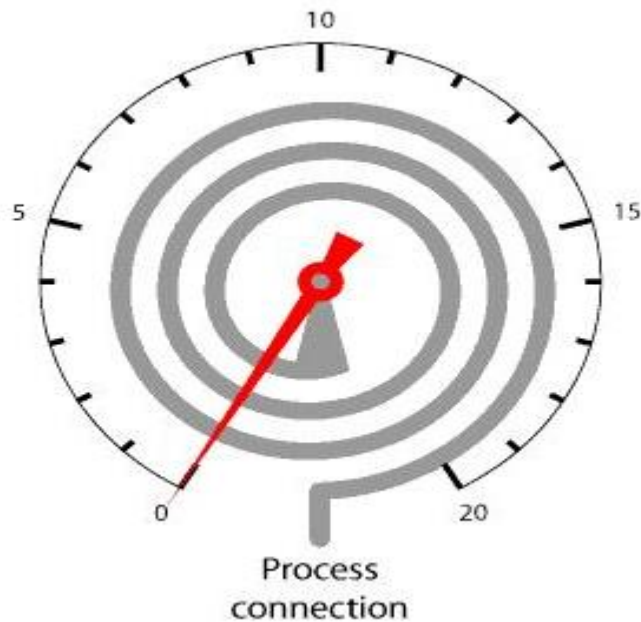
- One end of the tube is sealed or closed and other end is opened to enter the fluid.
- As pressurized fluid is entered, tube free end tends to straighten out on account of the pressure applied.
- Due to straightening of free end displacement occurs which is further amplified through mechanical linkage.
- The amplified displacement of free end may be used to move a pointer on the scale calibrated in terms of pressure or applied to a electrical displacement transducer.
- Bourdon tubes are made of brass, alloy steel, stainless steel, bronze, phosphor bronze, beryllium copper etc.



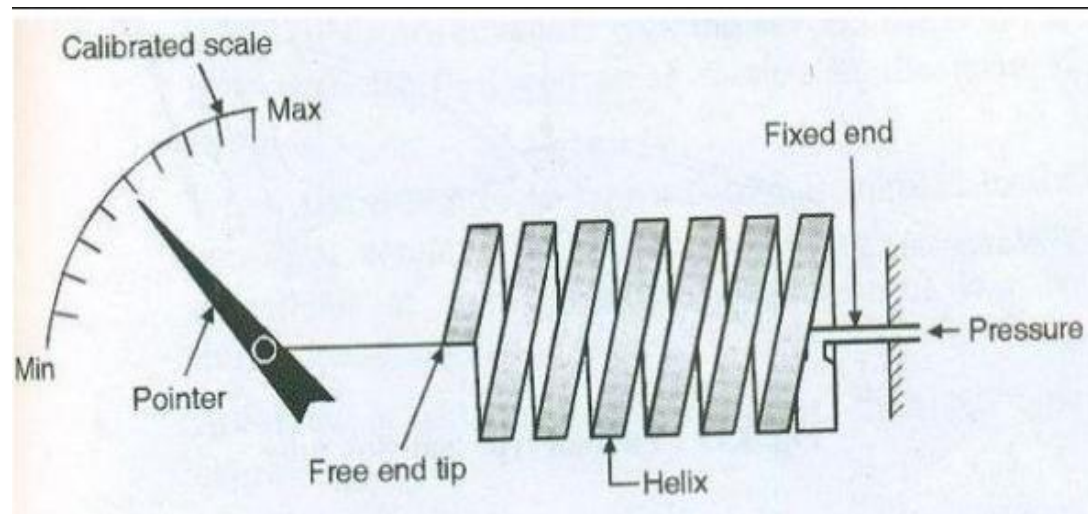
Bourdon Tube Pressure Gauge

Spiral or Helix Type Bourdon Tube

- Displacement of free end varies inversely as the wall thickness and depends upon the cross-sectional of tube.
- Displacement varies directly to the arc length.



Spiral Type Bourdon Tube

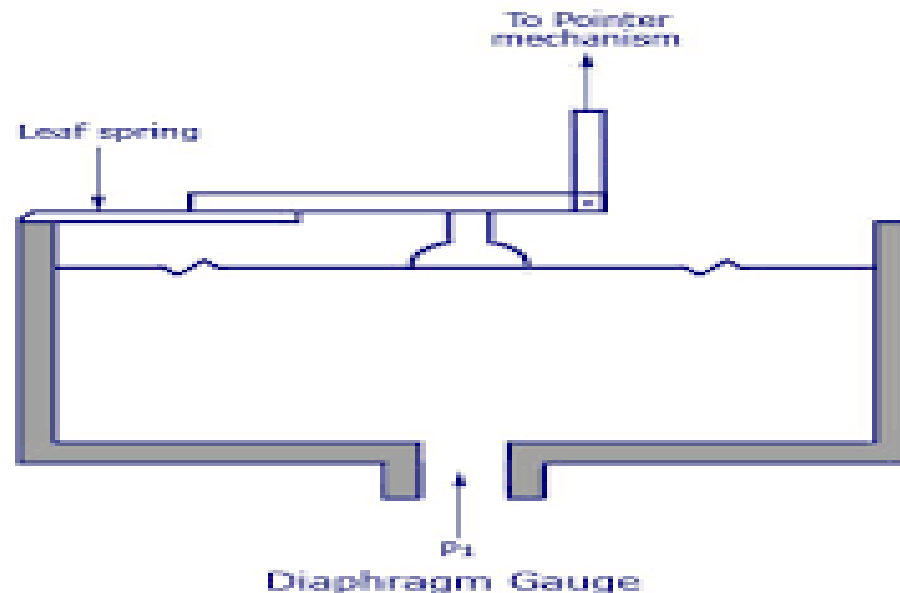


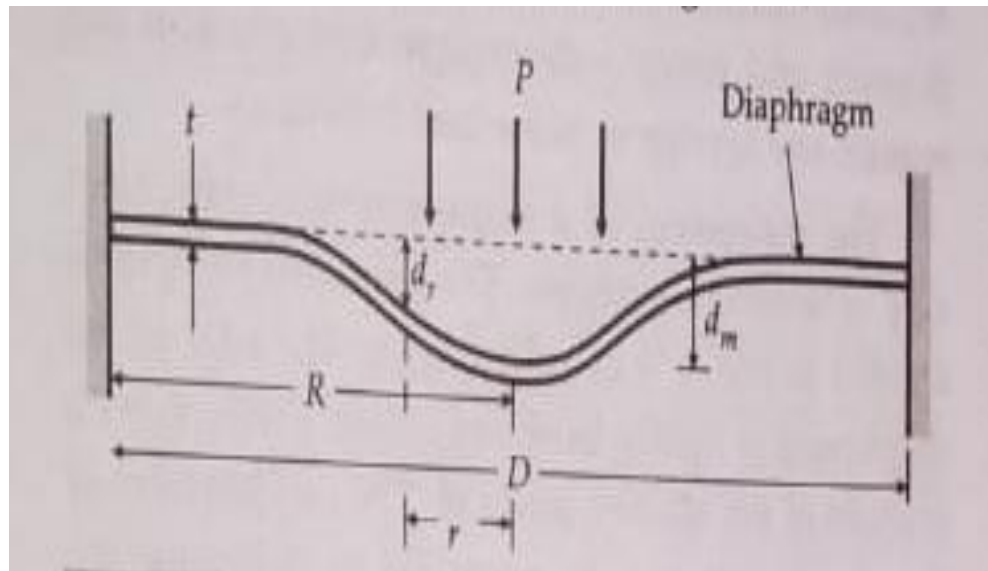
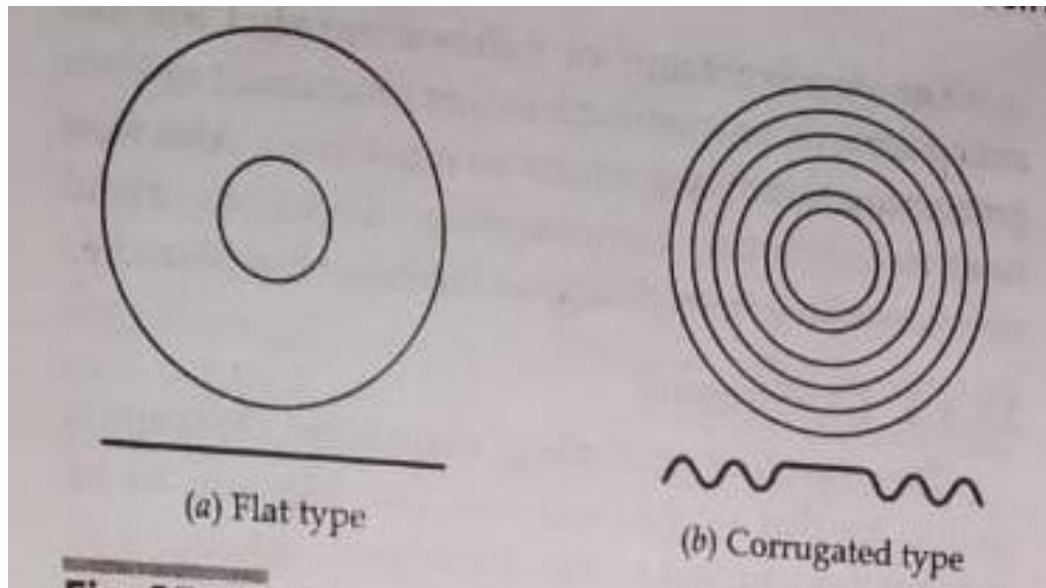
Helix Type Bourdon Tube

- Arc length can be increased by spiral or Helix type bourdon tube.
- In spiral or Helix bourdon, more displacement of the free end can be obtained by increasing the number of turns so further amplification is not required.
- As pressure is applied, it tends to uncoil producing relatively long displacement.
- Accuracy and range of spiral bourdon is higher compared to C-type element.
- The number of coils employed in helix elements depends upon the pressure to be measured.
- Minimum three coils and maximum sixteen coils are used in Helix type Bourdon tube.
- Pressure range is from 10-20,000 kpa.

Diaphragm

- Diaphragm is made of either thin membrane or circular plates.
- Thin membrane or circular plates are clamped around the circumference between two solid rings.
- Pressure to be measured is applied at the centre of Diaphragm.
- Deflection in Diaphragm is proportional to the applied pressure.
- Movement or deflection depends on its thickness and diameter.
- Thin membrane or circular plates may be one plate of capacitor and electrical output may be obtained.



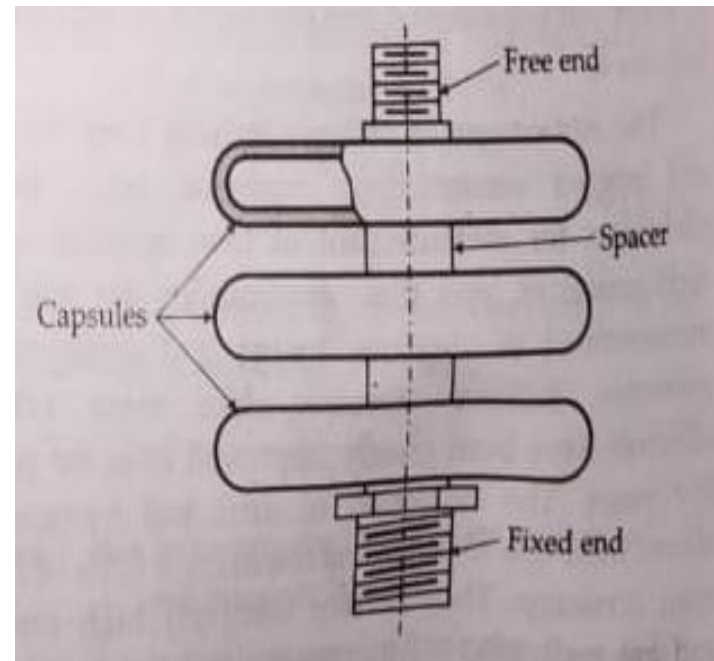
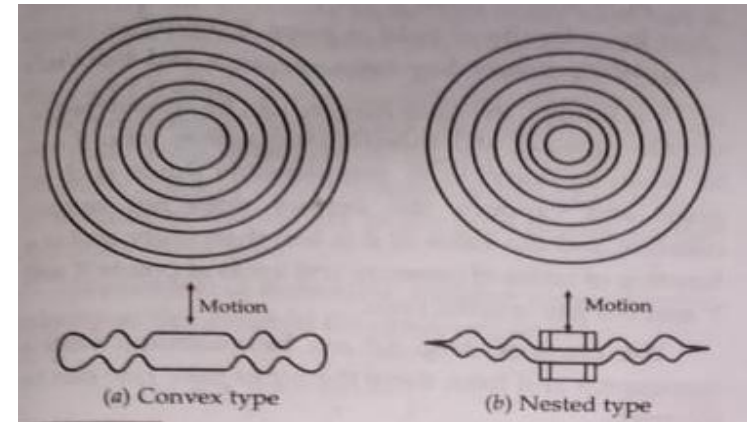


$$\text{Pressure } P = \frac{256 E t^3 d_m}{3(1-\nu^2) D^4} \text{ N/m}^2$$

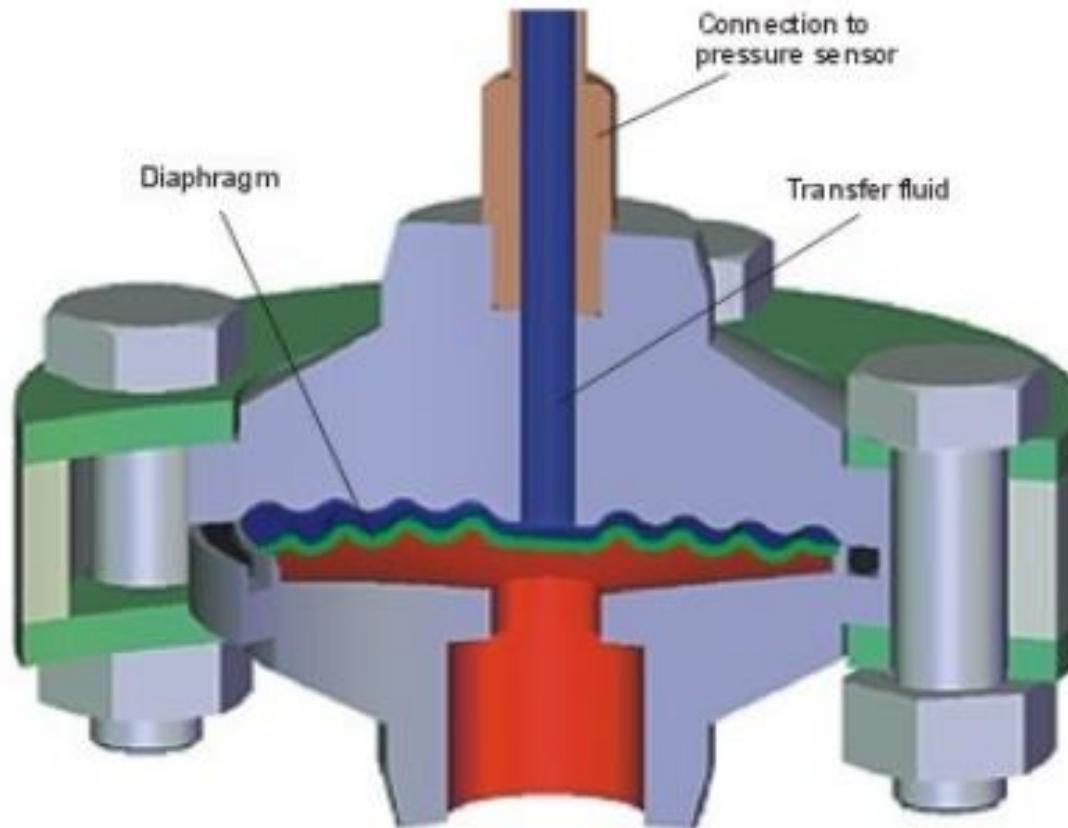
where E = Young's modulus ; N/m^2 ,
 t = thickness of diaphragm ; m,
 D = diameter of diaphragm ; m,
 R = radius of diaphragm ; m,
 ν = Poisson's ratio,
 and d_m = deflection at the centre of the diaphragm ; m

Capsule Type Diaphragm

- Two diaphragms are bonded together at their circumferences by soldering or pressure welding to form a capsule.
- Convex and nested type arrangements are used for capsule elements.
- A diaphragm element may consist of one capsule or two or more capsules axially connected to each other.
- Total deflection is the sum of the deflection of individual capsules.
- Diaphragms are made of stainless steel, phosphor bronze, beryllium copper, Monel, Nickel etc.
- It is used for low pressure including vacuum.
- The range of operation is 0-200 kN/m²



Pictorial View of Diaphragm



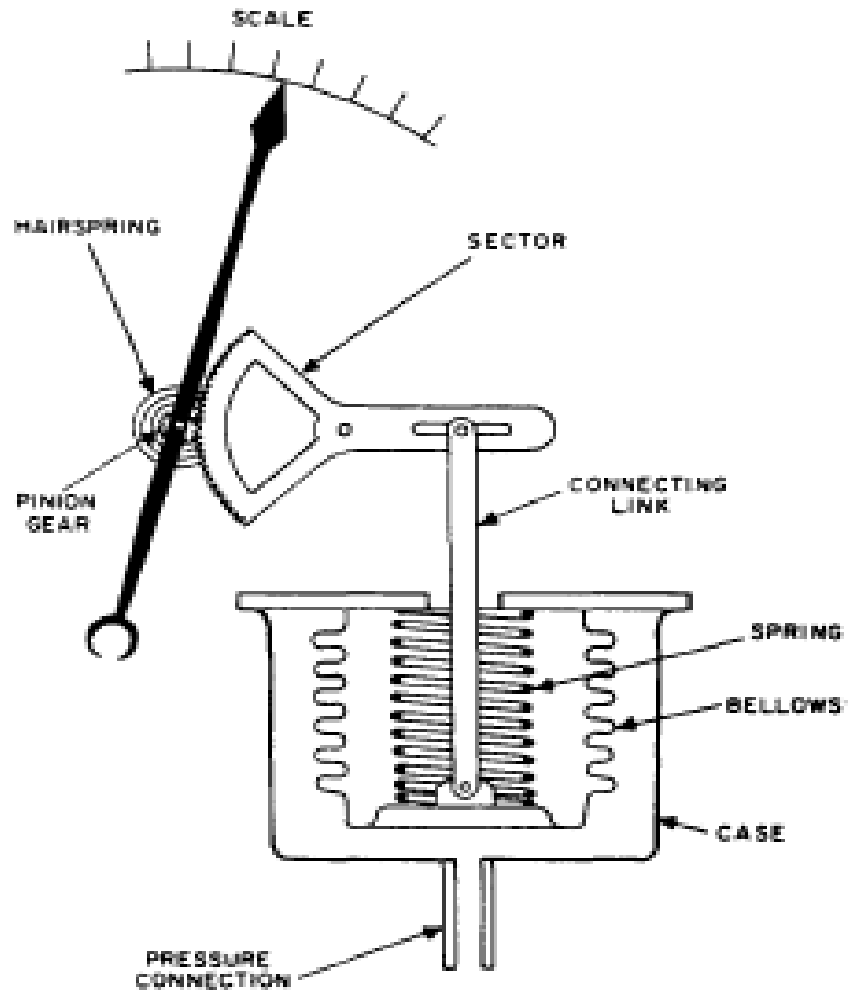
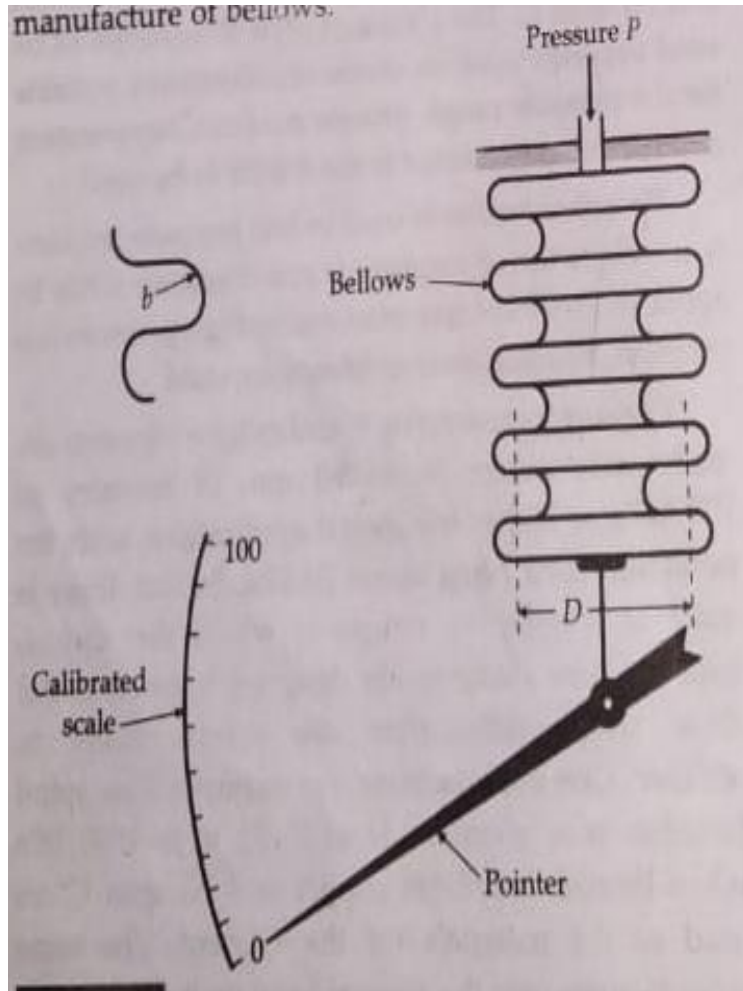
Bellows

- A metallic bellow is a series of circular parts.
- These parts are formed such that they expand or contract axially by change in pressure.
- The metals used in construction of bellow must be flexible, ductile enough for easy fabrication and high resistance to fatigue failure.
- Materials used are Brass, Bronze, Beryllium Copper, alloy of Nickle and Copper, Steel and Monel.
- A spring is inserted within the bellows which brings the bellow in initial position after removal of pressure.
- The Displacement of Bellow element is given by:

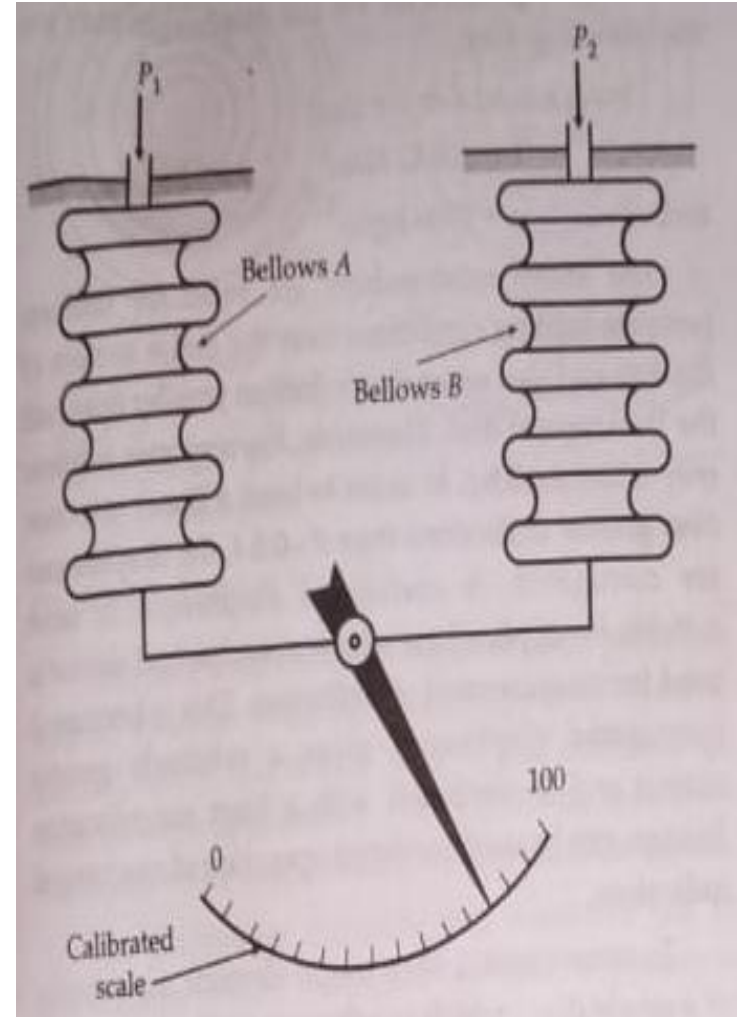
$$d = \frac{0.453 P_b n D^2 \sqrt{1 - \nu^2}}{E t^3}$$

Bellows

manufacture of bellows.

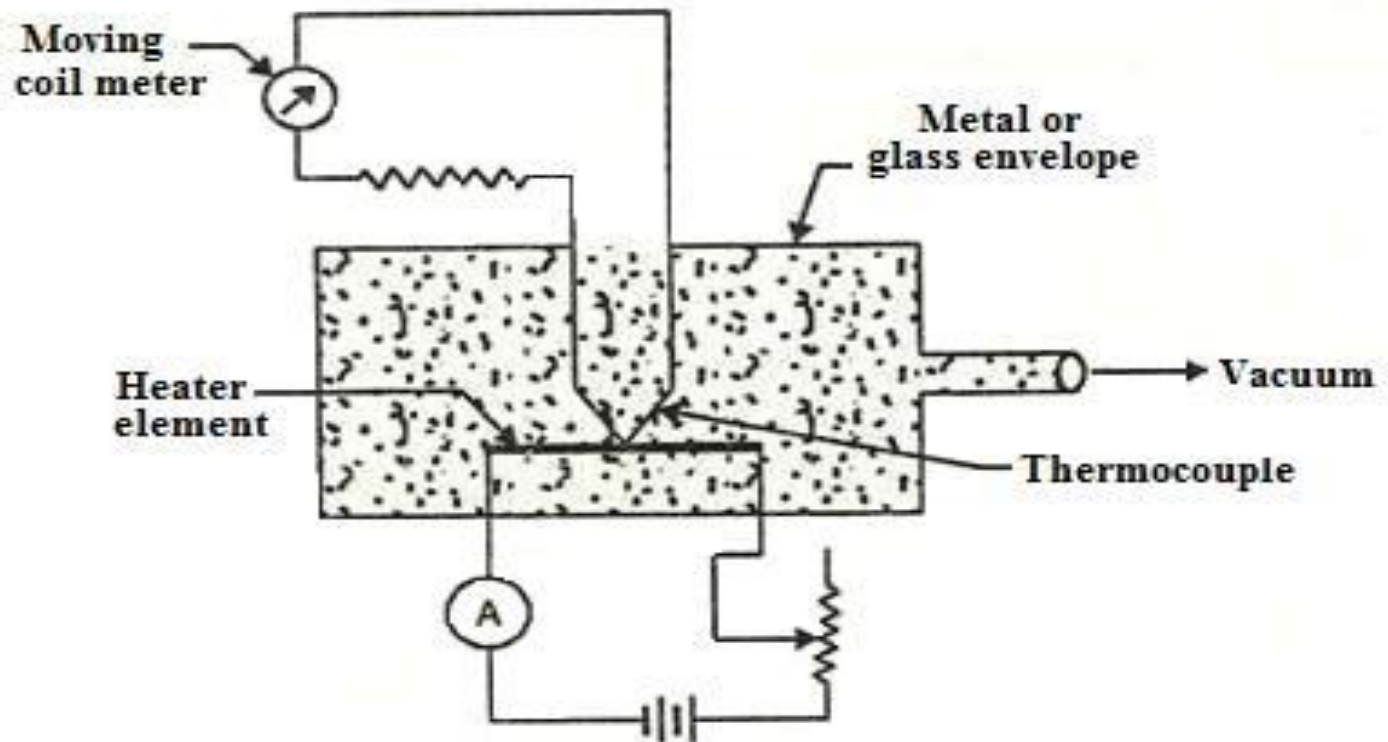


- Absolute, Gauge and differential pressures are measured with the help of two bellows.
- There are two bellows A and B.
- The pressure applied to Bellow A is P_1 and Bellow B is P_2 .
- In case to measure the absolute pressure, Bellow B is evacuated and resultant pressure P_1 is the absolute pressure.
- In case gauge pressure, Bellow B is opened up to atmosphere with pressure P_2 is equal to the atmospheric pressure and therefore the reading of gauge is gauge pressure.
- Differential pressure $P = P_1 - P_2$ is done when pressure P_1 is applied to Bellow A and pressure P_2 is applied to Bellow B.



Thermocouple Vacuum Gauge

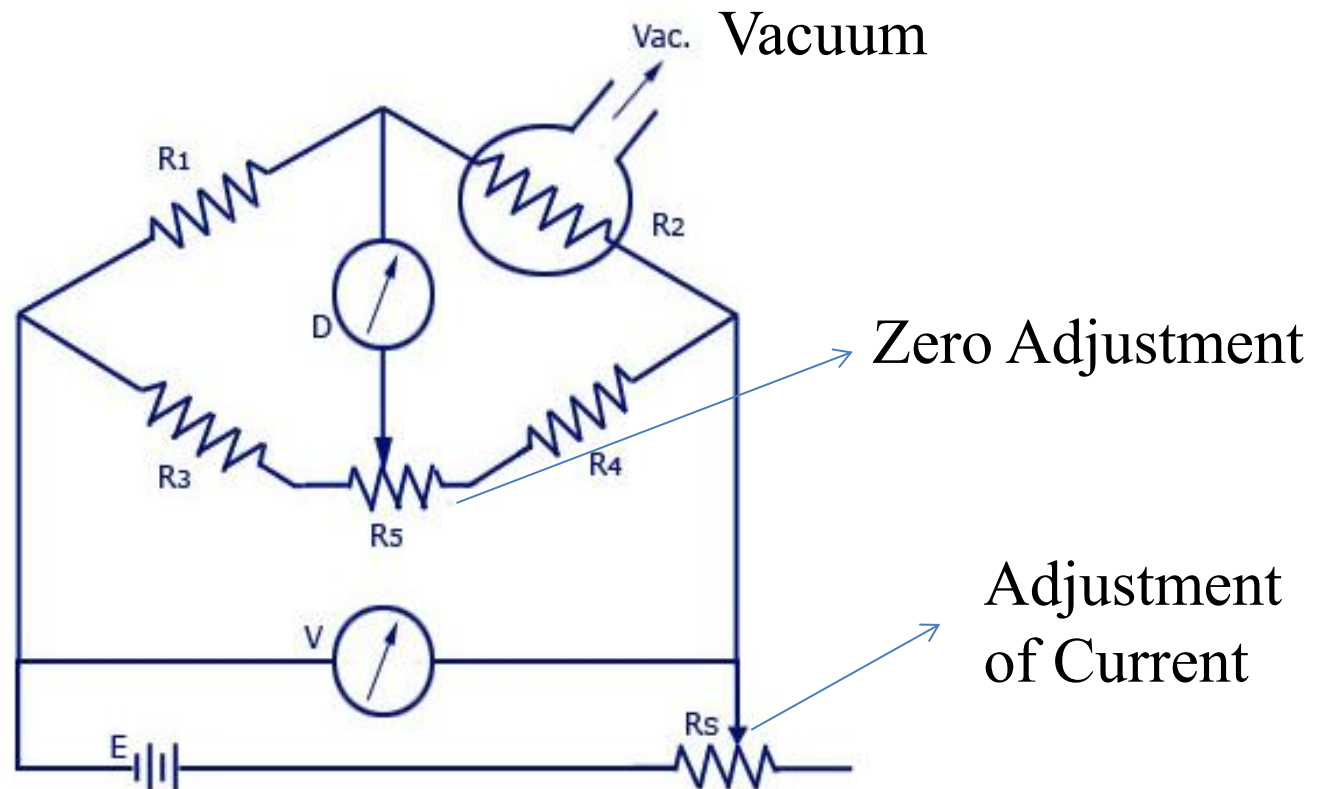
- This is used to measure vacuum pressure.
- This gauge operates on the principle that at low pressure the thermal conductivity of a gas (Argon, Helium and Hydrogen) is function of pressure.



- This gauge consists of a heater element (heated a temperature 50° to 400°C by a constant current).
- The heater element and thermocouple are enclosed in a glass or metal envelop which is vacuumed.
- Temperature of heater depends upon the amount of heat lost to the surroundings by conduction and convection.
- At pressure below 10^{-3} mm Hg, temperature of heater wire is a function of pressure.
- Thermocouple provides an output voltage which is a function of temperature of heater element and consequently of the pressure of the surrounding gas.
- The moving coil instrument is directly calibrated to read the pressure.

Pirani Gauge

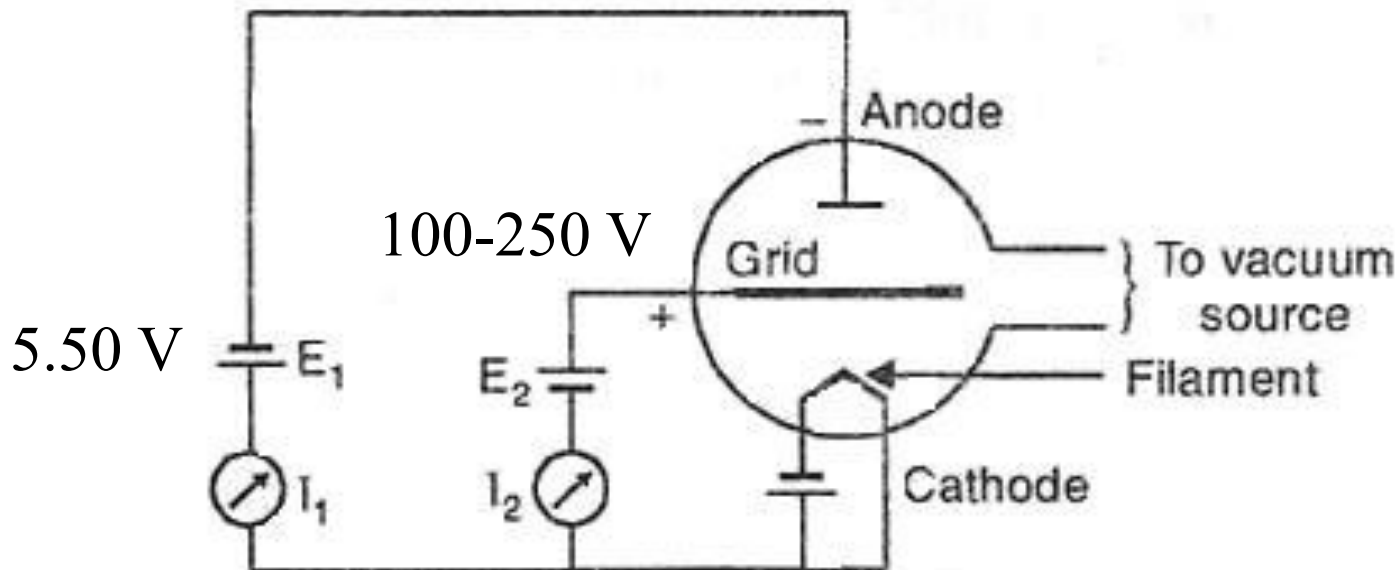
- The operation of a Pirani gauge depends on variation of the '**thermal conductivity**' of a gas with pressure.
- Below 1 mm of Hg pressure, thermal conductivity has linear relationship with pressure.



- Thermal conductivity of the gas is measured by detecting the amount of heat lost from an electrically heated wire placed in the gas.
- Heat is dissipated from the wire by conduction through the gas.
- Greater the thermal conductivity of the gas, lower will be the temperature of the heater wire.
- Electrical resistance varies with temperature so the resistance of the heater wire is a measure of the pressure.
- A Wheatstone bridge is used to measure the resistance of the heater wire.
- Milli-ammeter is used to measure current on account of unbalance in the bridge and is calibrated in terms of pressure.
- Pirani gauge is useful for pressure ranging from 10^{-1} to 10^{-3} mm of Hg.

Ionization Type Vacuum Gauge

- Ionization is a process of removing electron from atom and producing free electron and positively charged ion.
- Ionization may be done by the collision of a high speed electrons from the atom.
- Electrons are emitted from the heated cathode.



- Emitted electrons are accelerated towards the grid which is positively charged.
- Some of the electrons are captured by the grid and producing grid current I_G .
- Electrons having high kinetic energy pass through and causes ionization of gas atoms.
- The positive ions so produced are attracted to plate which is at negative potential and current I_p is produced in the plate.
- It is found that the pressure of gas is proportional to ratio of plate to grid current.

$$P = \frac{I_P}{S I_G}$$

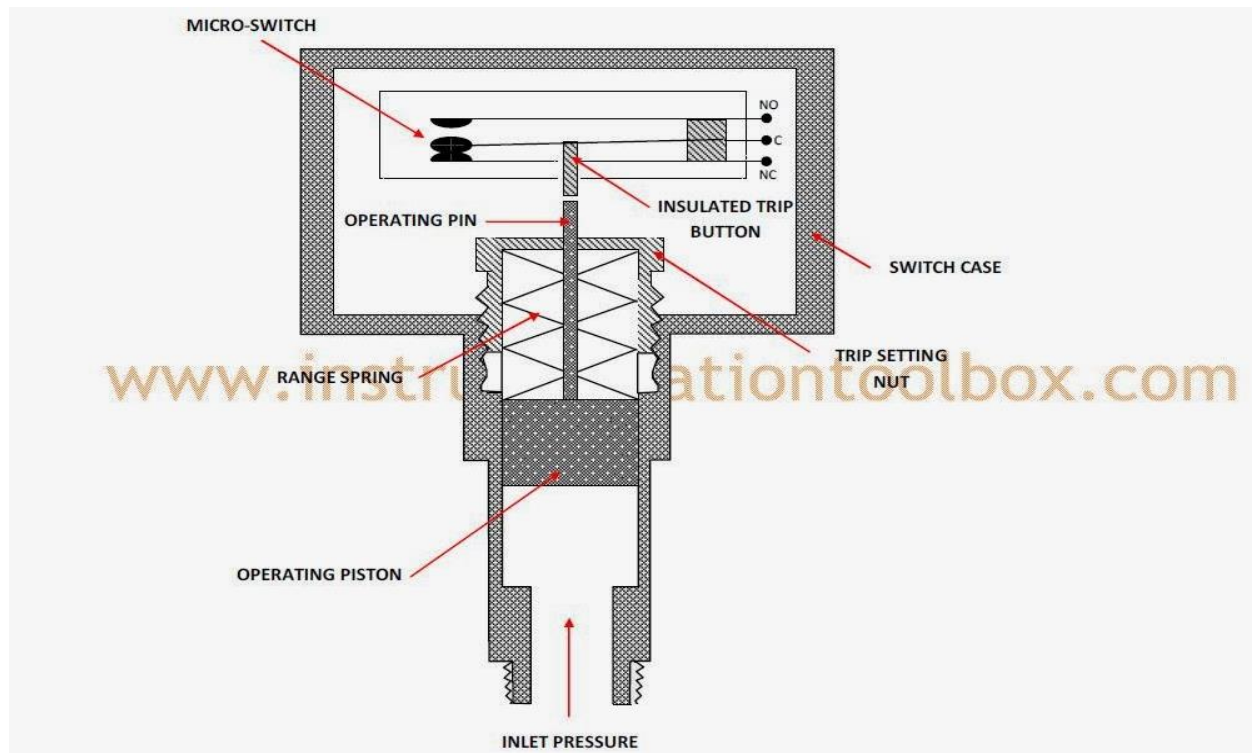
– S is sensitivity of the gauge.

- Pressure that can be measured by ionization gauge range from 10^{-3} to 10^{-8} mm of Hg.

Pressure switch

- A **pressure switch** is a form of switch that closes an electrical contact when a set fluid pressure has been reached on its input.
- The switch may be designed to make contact either ‘on’ as pressure rises or ‘on’ as pressure falls.
- Pressure switches are widely used in industry to automatically supervised system and control systems that use pressurized fluids.
- There are two basic types of pressure switches: **Solid state** and **Mechanical**.
- A solid state switch uses a metal diaphragm about the size of a dime (1.791 cm) with a micro-miniature strain gauge etched onto its dry surface.

- As pressure changes the impedance of the strain gauge is also changed.
- A digital comparator monitors the output from the strain gauge and compares it to the user specified set point.
- When the set point is reached, the electronic circuit turns on a solid state relay and completing the circuit.



Sectional View of Pressure Switch

- **Micro-Switch:** This switch is used to make or brake the electric circuit when pressure switch operates.
- **Insulated Trip Button:** This button causes the NO or NC contacts to the switch when pressure switch actuates.
- **Operating Pin:** Operating pin is attached to the operating piston. As inlet pressure changes, either operating pin moves up and make contact to the trip button or moves down to brake contact to the trip button.
- **Trip Setting Nut:** It is used to adjust the set point of the pressure switch.
- **Range Spring:** This is pre-compressed spring determines the range of operation.
- **Operating Piston:** This is part of pressure switch which is directly connected with process.
- **Switch Case:** The enclosure containing micro-switch and other accessories of the pressure switch.

Mechanical Pressure Switch

- **Diaphragm switch:** It employs a weld-sealed metal diaphragm to activate the switch. They have an operating pressure up to 150 psi and an accuracy to $\pm 0.5\%$.
- **Bourdon tube switch:** It uses a weld-sealed bourdon tube for switch actuation. Operating pressure ranges from 50 to 18,000 psi and accuracy is to $\pm 0.5\%$.
- **Diaphragm piston switch:** This switch is equipped with a diaphragm which acts on a piston. The piston then actuates the switch. Their operating pressure is from vacuum to 1600 psi and accuracy is to $\pm 0.2\%$.
- **Piston switch:** It employs a piston to directly activate the switch. Their operating pressure is to 12,000 psi and their accuracy is to $\pm 0.2\%$.
- **Differential pressure switch:** It consists of two diaphragms, one for high and low process pressure. The sensor detects the pressure difference between two sources and actuates the switch.

Pressure Switch Setting

- A mechanical Bourdon tube switch with a range of 2 to 60 psi has a minimum dead-band of 3 psi.
- There is a big difference in field adjustment setting.
- Mechanical switches typically require screw drivers or wrenches and setting the switch with watching on dial by trial and error method.
- The solid-state A-B switch with the same range has a minimum Dead-Band of 0.30 psi, a ten-fold improvement in performance.
- The solid state switch is set by pushing a few buttons and programming in the exact set point.

Applications of Pressure Switches

- Switch a household well water pump automatically when water is drawn from the pressure tank.
- Switching off an electrically driven gas compressor when a set pressure is achieved in the reservoir.
- Charge control inside a battery.
- Activating an alarm if aircraft cabin pressure drops.
- Pressure panels on sliding doors.
- Multiple pressure transducers can be attached to pipes to detect a break or leak in a pipe. The precise location of the leak is indicated by the transducer that is showing the largest drop in pressure

Basics of Piping and Instrumentation Diagram (P & ID)

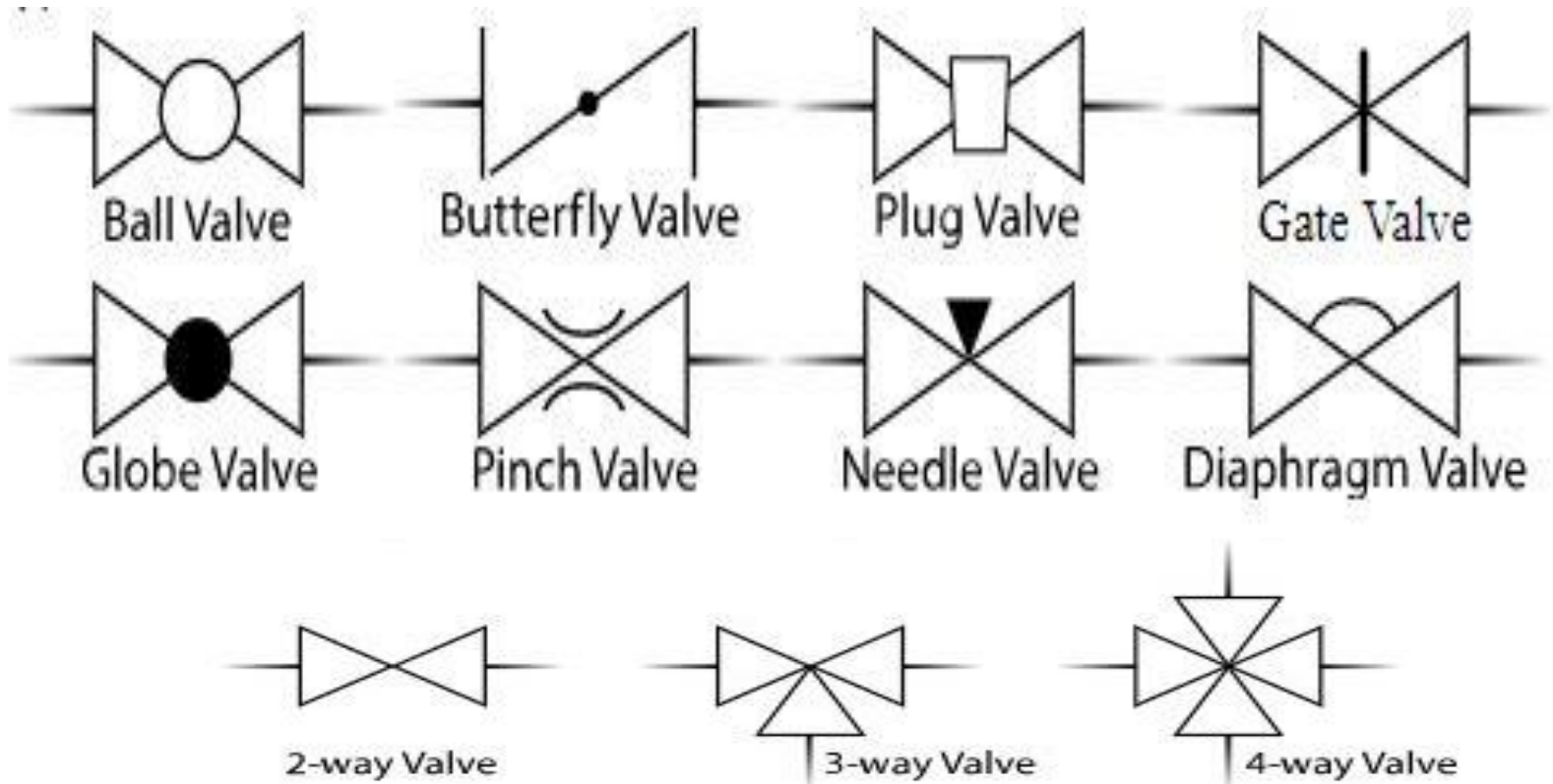
- A piping and instrumentation diagram (P&ID) is a graphic representation of a process system that includes the piping, vessels, control valves, instrumentation and other process components and equipment in the system.
- The P&ID is the primary schematic drawing used for laying out a process control system's installation
- The P&ID is used in Oil, Gas and Chemical Process Industries.

Phases of Use for P&IDs

- Design and layout of process system.
- Component specification.
- Installation of equipments and develop of control system schemes.
- Safety and operational analysis (HAZOP – hazard and operability study).
- Installation and/or build-out of the system.
- Startup, shutdown and operating schemes and procedures.
- Employee training of process system operation.
- Maintenance and modification to the system.

Symbols Used in P&IDs

- There are standard symbols used to represent the components in these diagrams.
- Valve Symbols for P&IDs



Process Lines

Process flow line



Instrument supply
or process connection
(impulse line)



Waveguide



Undefined



Pneumatic signal
(continuous)



Pneumatic signal
(discrete – on/off)



Capillary tube



Hydraulic signal



Electric signal
(continuous)



(or)



Electric signal
(discrete – on/off)



(or)



Fieldbus network



Data link
(smart instrument)



Mechanical link



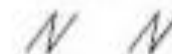
Data link
(common system)



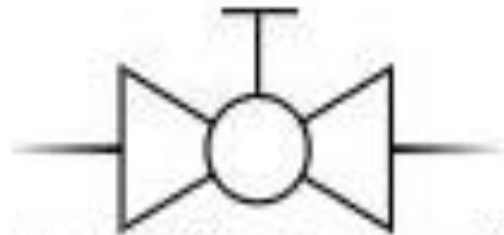
Data link
(independent systems)



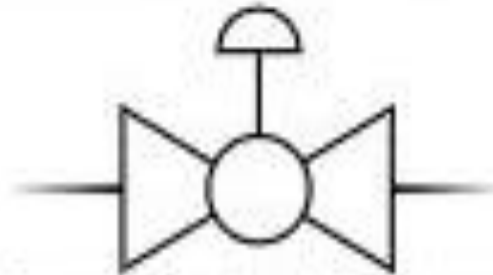
Radio link



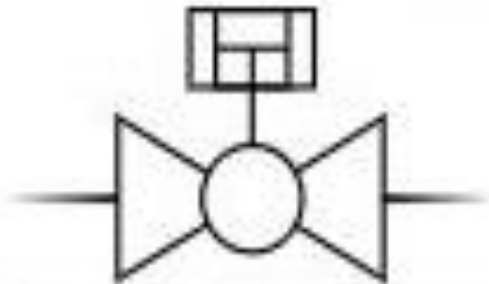
Type of Actuators



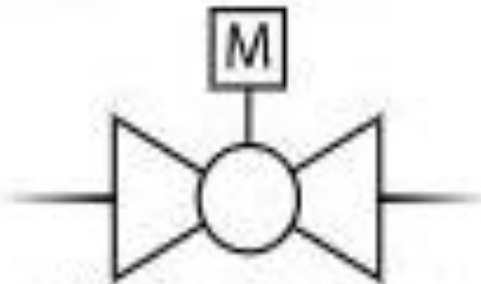
Manually Operated



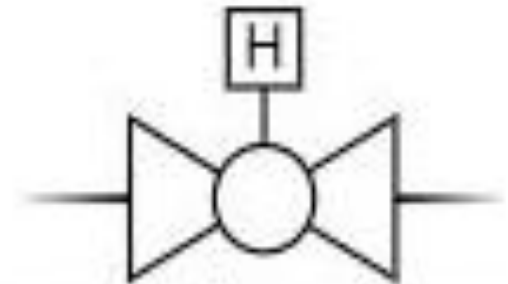
Pneumatic Actuator
(Diaphragm Type)



Pneumatic Actuator
(Rotary Piston Type)



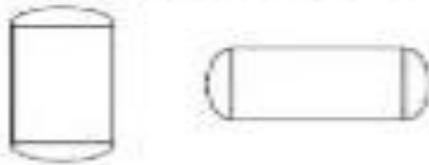
Electric Actuator



Hydraulic Actuator

Process Equipments

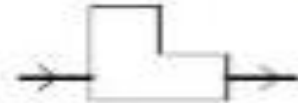
Pressure vessels



Centrifugal pump



Positive-displacement pump



Single-stage reciprocating compressor



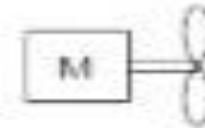
Dual-stage reciprocating compressor



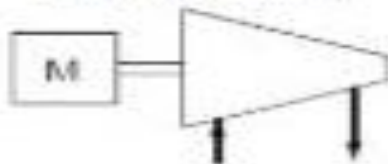
Rotary screw compressor



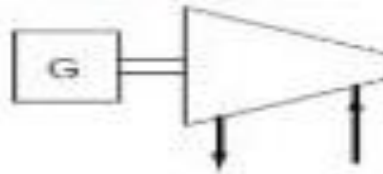
Motor-driven fan



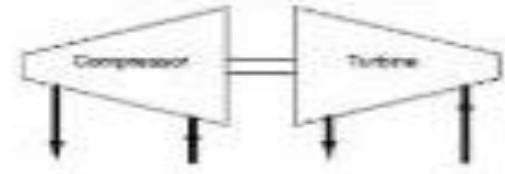
Motor-driven axial compressor



Turbogenerator



Turbocompressor



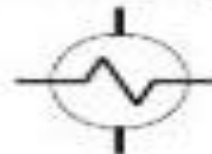
Mixer



Conveyor belt



Shell-and-tube heat exchanger

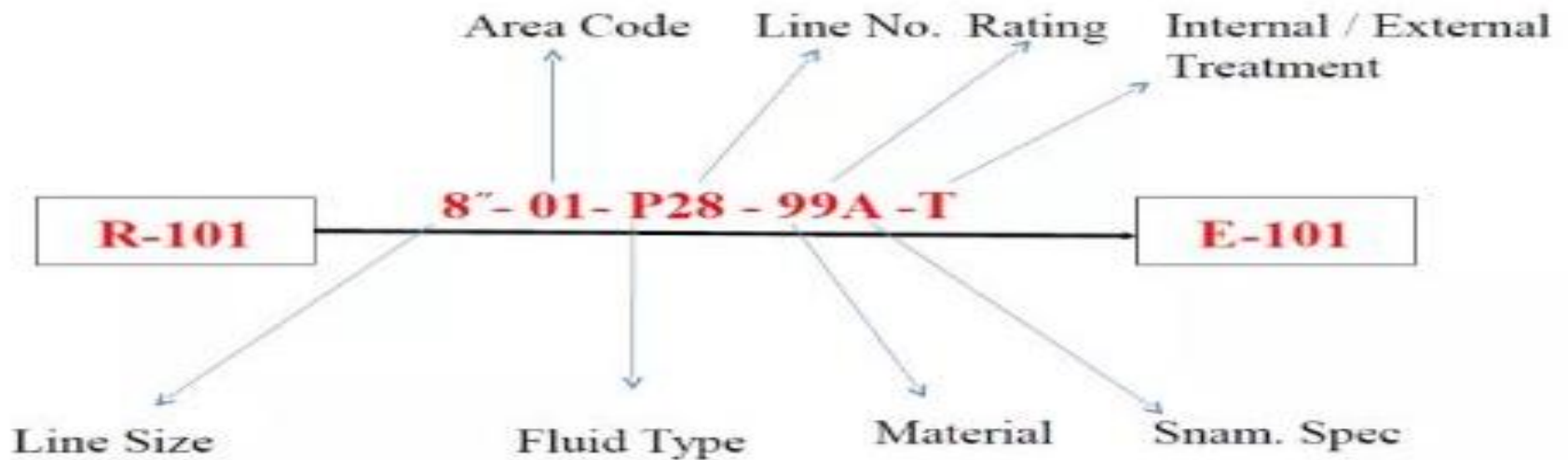


Jacketed vessel



How to read a P & ID

- Instruments used in the process and their connections are drawn in a P & I diagram.
- The process is simply described in the diagram.
- Specifications of the instruments are written on the instrument symbol.
- The specifications of the area and line are written on the P & I diagram.

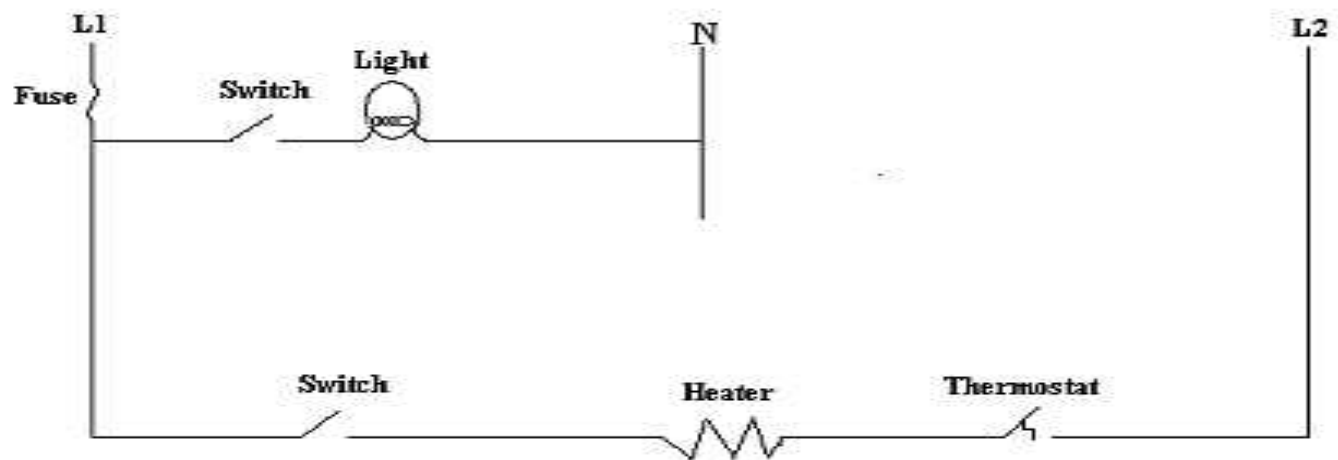


Basic Electrical and Electronics Drawing

- There are different types of diagram:
 - Wiring Diagram.
 - Block Diagram.
 - Circuit Diagram.

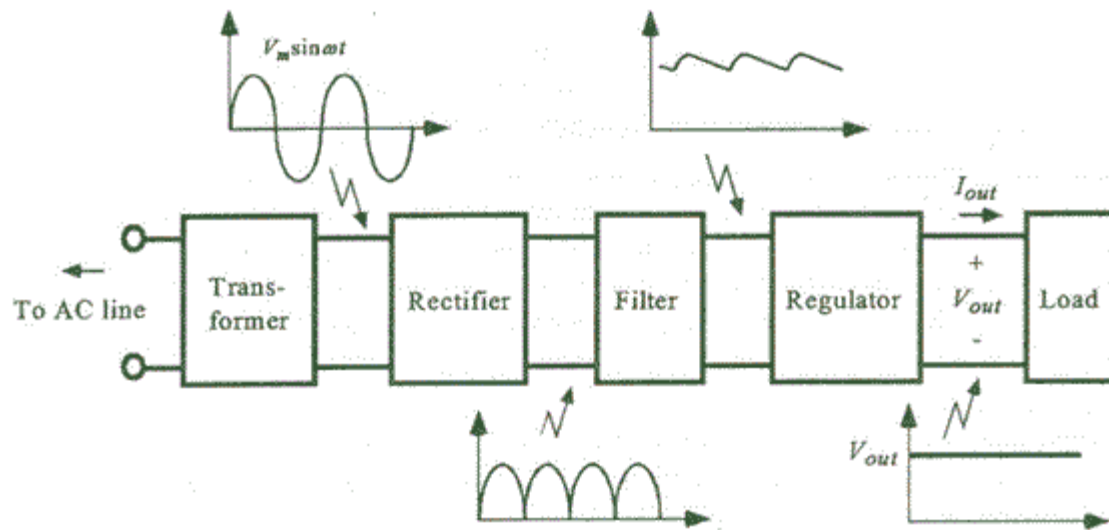
Wiring Diagram.

- Wiring diagram is a simple visual representation of the physical connections and physical layout of an electrical system or circuit.
- It shows how the electrical wires are interconnected and can also show where fixtures and components may be connected to the system.
- Wiring diagrams assist in building or manufacturing the circuit or electronic device.
- They are also useful for making repairs.



Block Diagram

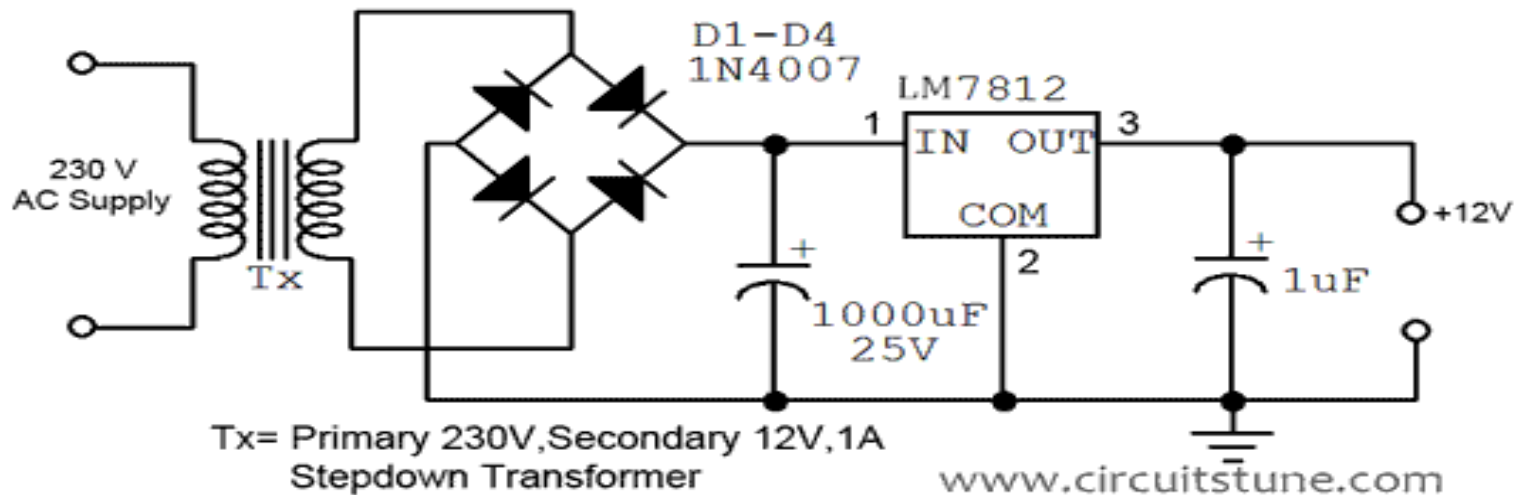
- A **block diagram** is a diagram of a system in which the principal parts or functions are represented by blocks connected by lines.
- A block diagram is a visual representation of a system that uses simple and labelled blocks.
- Block diagrams are a generalized representation of a concept and are not intended to display complete information in regards to design or manufacture.



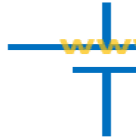
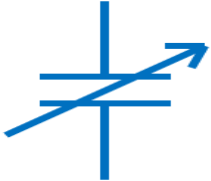
Components of a typical linear power supply

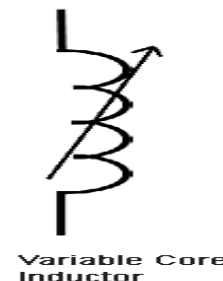
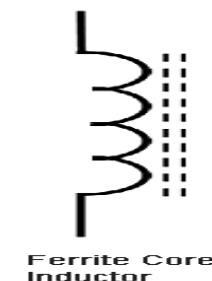
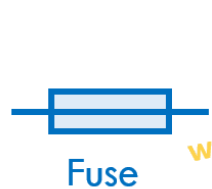
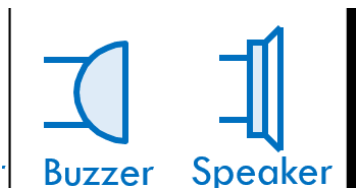
Circuit Diagram

- A **circuit diagram** (also known as an **electrical diagram**, **elementary diagram**, or **electronic schematic**) is a simplified conventional graphical representation of an electrical circuit.
- Unlike a block diagram or layout diagram, a circuit diagram shows the actual circuits being used.

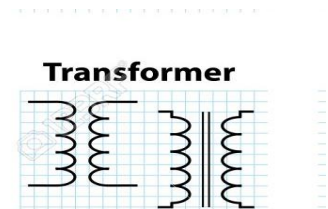


Symbols

 Cell	 Battery	 AC Voltage	 AC Current	 Lamp
 Capacitor	 Variable C	 SPST  SPDT	 Wires not crossing	 Wires crossing



 Source	 Ideal source	 Resistor	 Alternate resistor	 Variable resistor
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— E — E —

Main Supply Line



Main Control Board



Distribution Board



Switch Board

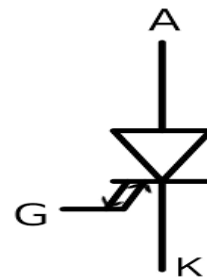
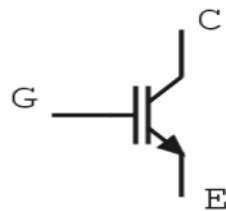
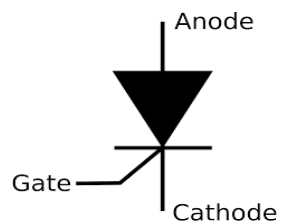
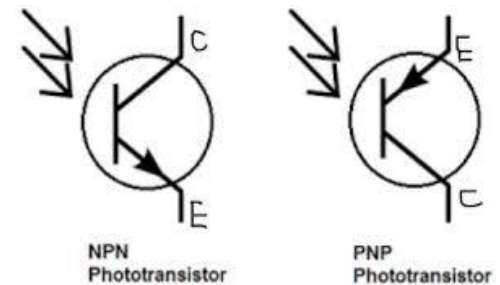
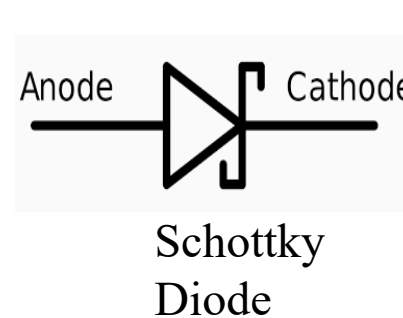
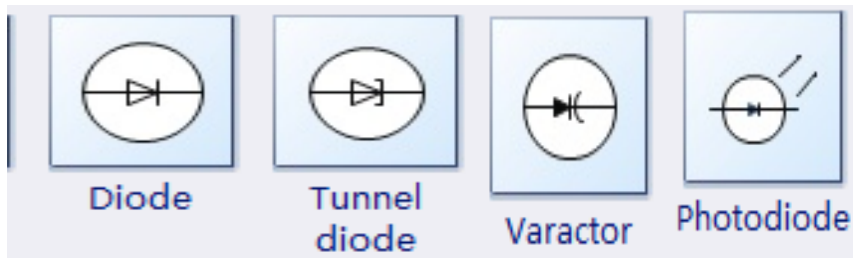
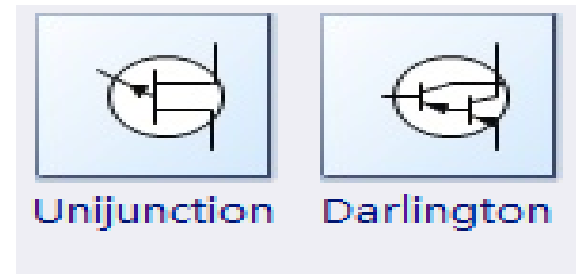
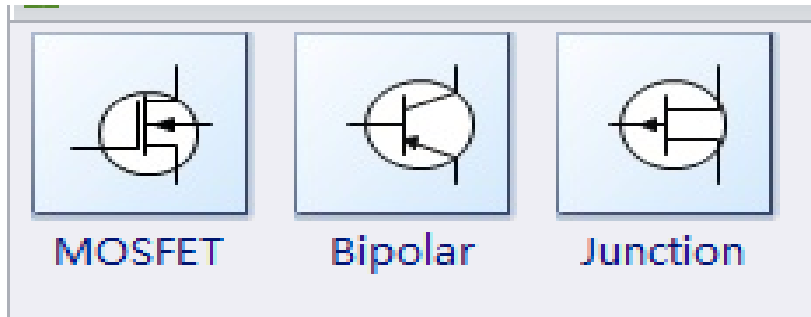


Electrical Outlet – 5 Amps.



Electrical Outlet – 15 Amps.

Electronics Devices



Thank you

Why mercury is used as standard liquid in pressure measurement

- Mercury has high density so small column size is required.
- If water is used in pressure measurement than 13.5 times column size is required.
- It does not evaporated like water.
- It has lower freezing point means it will not freeze upto 0° C.
- It is not stick on the glass.
- Is the only metal that is liquid at room temperature.
- **Mercury** expands and contracts evenly with temperature and **pressure** changes.
- $760 \text{ mm of Hg} = 1.033 \text{ Kg/cm}^2$
- $1 \text{ Pascal} = 0.0075 \text{ mm of Hg}$
- $1 \text{ atm} = 760 \text{ torr} = 14.7 \text{ psi}$